

PEMM — Prediction Error Measurement Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

POFPS — Prediction Outcome & Performance Standard

Operational Layer: Prediction Outcome & Predictive Performance

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Error Measurement

Version: 1.0

Status: Canonical · Open Module

Effective Date: 11 September 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

PEMM governs how differences between a Governed Prediction and its sufficiently governed Outcome Reference are measured, classified, normalized, qualified, and preserved after POCP has established a legitimate Prediction Outcome Comparison.

It governs the transition:

Governed Prediction Outcome Comparison → Governed Prediction Error Measurement

PEMM determines:

whether measurable error exists; what dimension contains the error; how the error may legitimately be quantified; what measurement method applies; what reference scale applies; whether normalization is required; whether multiple errors must remain separate; how timing, magnitude, range, categorical,



directional, threshold, and composite error are represented; how uncertainty and intervention affect interpretation; how error remains historically traceable.

Its foundational principle is:

Error measures the distance between prediction and outcome. It does not by itself explain why that distance exists.

Canonical Definition

The Prediction Error Measurement Module (PEMM) defines the operational governance requirements by which differences identified through a Governed Prediction Outcome Comparison are transformed into explicit, reproducible, appropriately scaled, dimension-specific, and historically traceable Prediction Error Measurements.

PEMM governs:

Error Eligibility; Error Object; Error Dimension; Error Direction; Error Magnitude; Absolute Error; Signed Error; Relative Error; Percentage Error; Range Error; Threshold Error; Timing Error; Duration Error; categorical error; directional error; composite error; normalized error; scale-sensitive error; asymmetric error; weighted error; Error Measurement Method; Error Qualification; Error State; Error Profile; Error History.

PEMM does not determine whether the predictor is good, bad, skilled, calibrated, reliable, or methodologically sound.

Operational Role

PEMM is the third operational module within POFPS.

Its focused operational question is:

Where a governed comparison identifies a difference between prediction and outcome, how should that difference be measured without confusing numerical distance with predictive quality, methodological validity, or causal explanation?

PEMM receives:

Governed Prediction Outcome Comparison
from POCM.

It produces:

Governed Prediction Error Measurement
for subsequent use by:

PCALM — Prediction Calibration Module
and:

PPCM — Predictive Performance Module
where applicable.

Internal Module Flow

Prediction Outcome Association → Prediction Outcome Comparison → Prediction Error Measurement → Prediction Calibration → Predictive Performance

Current Position:

Prediction Error Measurement

1. Prediction Error

Prediction Error is the governed difference between a prediction and the corresponding Outcome Reference across one or more measurable predictive dimensions.

Conceptually:

Prediction State – Outcome State → Prediction Error

The exact mathematical direction may vary by method.

The governance meaning must remain explicit.

2. Error Eligibility

Not every Prediction Outcome Comparison produces a meaningful numerical error.

Error may be:

Measurable

Measurable with Qualification

Partially Measurable

Categorically Representable

Not Meaningfully Measurable

Unable to Determine

3. Error Requires Comparison

PEMM must not independently select the Outcome Reference or redefine the comparison.

POAM owns association.

POCM owns comparison.

PEMM measures the difference established through that governed comparison.

No Governed Comparison ≠ Governed Error Measurement

4. Error Object

The Error Object identifies what predictive property is being measured.

Possible Error Objects include:

value; magnitude; direction; range; threshold; timing; duration; category; probability-related result; composite prediction component.

5. Error Dimension

A prediction may contain multiple independent error dimensions.

Example:

Prediction:

Revenue grows 15% by Q2.

Outcome:

Revenue grows 6% by Q4.

Possible error dimensions include:

magnitude error; timing error.

These should not automatically collapse into one value.

6. Multi-Dimensional Error

A prediction may be:

directionally correct; numerically inaccurate; temporally inaccurate; geographically partial.

PEMM should preserve dimension-specific errors.

One Prediction $\neq$ One Error Value by Default

7. Absolute Error

For numerical predictions, Absolute Error may be represented conceptually as:

$|\text{Predicted Value} - \text{Outcome Value}|$

Example:

Predicted:

120

Outcome:

111

Absolute Error:

9

The unit must remain explicit.

8. Signed Error

Signed Error preserves the direction of numerical difference.

Conceptually:

$\text{Predicted Value} - \text{Outcome Value}$

Example:

Prediction:

120

Outcome:

111

Signed Error:

+9

The sign convention must be documented.

9. Overprediction

Where predicted value exceeds the governed outcome:

$\text{Predicted Value} > \text{Outcome Value}$

the prediction may be classified as:

Overprediction

where that terminology is appropriate.

10. Underprediction

Where predicted value falls below the governed outcome:

Predicted Value < Outcome Value

the prediction may be classified as:

Underprediction

11. Absolute vs Signed Error

Absolute Error measures distance.

Signed Error preserves direction.

Absolute Error $\neq$ Signed Error

Both may be necessary.

12. Relative Error

Relative Error expresses error relative to an appropriate reference value.

Conceptually:

Absolute Error / Reference Value

The denominator must be explicitly governed.

13. Percentage Error

Percentage Error may express relative error as a percentage.

However, percentage error can become unstable or misleading when the reference value is:
zero; near zero; negative; structurally inappropriate.

Therefore:

Percentage Error $\neq$ Universally Valid Error Metric

14. Zero-Denominator Problem

Where the outcome value equals zero, ordinary percentage-error calculations may be undefined.

PEMM must not manufacture a percentage merely to preserve metric consistency.

15. Near-Zero Distortion

A very small denominator may create extremely large percentage errors even when the absolute difference is operationally small.

The metric should therefore remain qualified.

16. Scale Context

An error of:

10 units

may be enormous on a scale of 20 and negligible on a scale of 10 million.

Therefore:

Same Absolute Error $\neq$ Same Predictive Significance

17. Normalized Error

Normalized Error may place prediction errors onto a common reference scale.

Normalization method must remain documented.

18. Normalization Basis

Possible normalization bases may include:

observed value; prediction range; historical range; domain scale; benchmark; operational tolerance.

The selected basis changes the interpretation.

19. Raw Error vs Normalized Error

A normalized metric should never silently replace the raw error.

Where practical, both should remain traceable.

Normalized Error $\neq$ Raw Error

20. Numerical Precision

Prediction and outcome precision should remain visible.

Prediction:

100

may not carry the same precision as:

100.000

PEMM should not manufacture false error precision.

21. Measurement Precision

Outcome measurements themselves may have finite precision.

Error measurement should not imply greater certainty than the Outcome Reference supports.

22. Rounding Error

Some apparent prediction error may arise solely from rounding.

Example:

Prediction:

4.2%

Outcome:

4.24%

Reported outcome:

4.2%

The system should preserve the relevant precision level.

23. Measurement Uncertainty

The Outcome Reference may contain measurement uncertainty.

PEMM should preserve this where material.

Observed Value $\neq$ Infinitely Precise Reality

24. Prediction Range Error

Where the prediction specifies a range, PEMM may measure the distance of the outcome from the nearest range boundary where the outcome falls outside the range.

Example:

Prediction:

10–15

Outcome:

18

Range Error:

3 above upper boundary

25. Within-Range Outcome

Where the outcome falls inside the predicted range:

Range Error = 0

may be appropriate for one metric.

But this does not mean the prediction was maximally informative.

A range of:

0–1,000

may contain the outcome while providing little predictive precision.

Within Range $\neq$ High Predictive Performance by Default

26. Range Width Context

Range Width should remain available to downstream performance assessment.

A narrow correct range and a broad correct range should not automatically be treated as equivalent.

27. Boundary Error

Where the outcome lies exactly at a predicted range boundary, the implementation should preserve whether boundaries were defined as inclusive or exclusive.

28. Threshold Error

For threshold predictions, error may measure distance from the threshold.

Example:

Prediction:

Inflation > 4%

Outcome:

3.2%

Threshold Difference:

–0.8 percentage points

29. Threshold Crossing

Where the threshold is crossed, simple threshold error may be zero for categorical resolution even though the observed value may be barely above or far above the threshold.

POCM and PPCM preserve the broader meaning.

30. Threshold Distance

Threshold Distance may preserve how far the Outcome Reference lies from the predicted threshold.

This may matter operationally even where categorical threshold resolution is identical.

31. Directional Error

Directional predictions may produce states such as:

Direction Correct

Direction Incorrect

Direction Neutral / No Change

Direction Unable to Determine

Directional error may be categorical rather than numerical.

32. Magnitude Error

Magnitude Error measures difference in predicted scale.

Example:

Predicted decline:

20%

Outcome decline:

7%

Magnitude Error:

13 percentage points

where appropriate.

33. Timing Error

Timing Error measures the difference between predicted timing and actual outcome timing.

Example:

Predicted event:

30 June

Actual event:

20 August

Timing Error:

51 days late

depending on the defined convention.

34. Early Timing Error

An event may occur earlier than predicted.

The error direction should remain visible.

Example:

–20 days

may represent 20 days earlier, if the convention is explicitly defined.

35. Late Timing Error

An event may occur later than predicted.

The timing difference should remain distinguishable from complete event non-occurrence.

36. Horizon Miss

If an event occurs outside the Prediction Horizon, PEMM may preserve:

event occurrence; Horizon Miss; timing distance beyond horizon.

Horizon Miss $\neq$ Event Non-Occurrence

37. Duration Error

Where duration is predicted:

Predicted Duration – Observed Duration

may be measured using the applicable time unit.

38. Sequence Error

Where order matters, Sequence Error may be represented through:

order mismatch; stage displacement; missing stage; reversed sequence.

A numerical score is not always necessary.

39. Categorical Error

Categorical predictions may require categorical error states rather than arbitrary numerical distance.

Example:

Predicted: Category A

Outcome: Category C

Unless a legitimate category-distance structure exists, assigning:

Error = 2

would be methodologically unjustified.

40. Category Distance

Numerical Category Distance may be used only where categories possess a legitimate ordered or metric relationship.

Different Category $\neq$ Numerical Distance by Default

41. Ordinal Error

Ordered categories may support ordinal difference.

Example:

Low $\rightarrow$ Medium $\rightarrow$ High $\rightarrow$ Critical

Predicted:

High

Outcome:

Critical

Ordinal Error:

one governed category level

where such representation is legitimate.

42. Binary Error

Binary predictions may produce:

Match

or:

Mismatch

But probabilistic predictions associated with binary outcomes should not be reduced to ordinary binary error alone.

43. Probability Error Boundary

If a prediction states:

70% probability

and the event occurs, the numerical difference between:

0.70

and:

1

is not by itself a complete predictive-performance measure.

Probability calibration requires repeated predictive context.

PCALM owns that space.

44. Probability Error vs Calibration

A probability prediction may support proper scoring methods or event-level probability loss measures where explicitly governed.

But:

Single Probability Error $\neq$ Calibration

45. Error Measurement Method

Every material Prediction Error Measurement should identify the method used.

Examples may include:

absolute error; signed error; relative error; percentage error; range deviation; timing deviation; categorical mismatch; domain-specific governed metric.

46. Method Selection

The Error Measurement Method should correspond to:

Prediction Object; measurement scale; outcome type; performance purpose.

Available Metric $\neq$ Appropriate Metric

47. Metric Appropriateness

A mathematically valid metric may still be methodologically inappropriate.

PEMM therefore governs not only calculation but metric suitability.

48. Metric Consistency

Comparable predictions should normally use consistent error methods unless a governed reason justifies change.

49. Metric Change

If the Error Measurement Method changes over time, performance history should preserve:
prior method; new method; effective time; reason.

50. Retrospective Metric Change

A metric should not be changed after outcomes become known simply because another method produces more favorable performance.

Outcome Knowledge $\neq$ Permission to Select a Better-Looking Error Metric

51. Error Transformation

Error may be transformed for:

normalization; aggregation; logarithmic representation; domain-specific scaling.

The transformation must remain traceable.

52. Error Sign Convention

Systems should explicitly define whether:

Prediction – Outcome

or:

Outcome – Prediction

determines the sign.

Otherwise, positive and negative errors may be misinterpreted.

53. Error Unit

Every quantitative error should preserve its unit.

Examples:

CHF; percentage points; days; MW; tonnes; degrees; units sold.

Error Number Without Unit $\neq$ Complete Error Representation

54. Percentage vs Percentage-Point Error

Prediction:

10%

Outcome:

7%

Difference:

3 percentage points

not necessarily:

3%

These must not be confused.

55. Currency Error

Financial prediction error should preserve:

currency; nominal or real basis; exchange-rate basis where applicable;
relevant reporting period.

56. Time-Scale Error

Timing errors should preserve the time unit and applicable calendar convention.

Examples:

calendar days; business days; months; quarters.

57. Measurement Transformation Record

Where POCM normalized units before comparison, PEMM should preserve that transformation reference.

58. Composite Error

Composite predictions may contain several error dimensions.

PEMM should produce a structured Error Profile before any aggregate score is considered.

59. Error Profile

An Error Profile may contain:

Direction Error: None

Magnitude Error: 12 percentage points

Timing Error: 3 months late

Range Error: 5 units above upper bound

This is often more informative than one aggregate number.

60. Aggregate Error

Multiple error dimensions may be aggregated only where a documented methodology justifies:
weighting; normalization; combination.

Multiple Errors $\neq$ Legitimate Single Error Score by Default

61. Error Weighting

Where errors are weighted, the weighting logic must remain explicit.

Example:

Timing may be more operationally important than magnitude in one prediction domain.

62. Weighting Before Outcome

Where possible, weights should be established before outcome knowledge.

This reduces retrospective optimization.

63. Retrospective Weight Manipulation

Changing weights after outcome knowledge to reduce measured error constitutes a governance risk.

64. Asymmetric Error

Overprediction and underprediction may have different operational consequences.

PEMM may preserve asymmetric error treatment where legitimately defined.

65. Asymmetric Loss Boundary

Operational consequence is not automatically the same as predictive error.

A 10-unit overprediction and 10-unit underprediction may have equal numerical error but different business consequences.

Prediction Error $\neq$ Operational Loss

66. Tolerance Band

Some prediction contexts may define an acceptable tolerance.

Example:

$\pm 2^{\circ}\text{C}$

or:

$\pm 5\%$

Tolerance should be defined before performance interpretation.

67. Within-Tolerance Error

An error may exist numerically while remaining inside an accepted tolerance.

Therefore:

Non-Zero Error $\neq$ Material Error by Default

68. Material Error

Materiality depends upon context.

PEMM may classify:

Negligible

Minor

Material

Major

Critical

where a governed materiality framework exists.

69. Error Materiality Boundary

PEMM may classify error magnitude or materiality.

It does not determine organizational consequence, liability, or decision quality.

70. Error Directionality

Repeated errors in the same direction may indicate systematic bias.

PEMM preserves error direction.

PPCM may later evaluate the performance pattern.

71. Systematic Error Candidate

Repeated overprediction or underprediction may generate:

Systematic Error Candidate

This is not yet proof of methodological bias.

72. Random Error Candidate

Alternating errors may appear more consistent with non-directional variation.

PEMM should not make unsupported causal conclusions.

73. Error Cause Boundary

PEMM measures error.

It does not automatically determine whether the error arose from:

methodology; data; assumptions; environment; model; human judgment; intervention.

Measured Error $\neq$ Established Error Cause

74. Error Attribution

Where another governed process establishes error attribution, PEMM may preserve the reference.

It should not manufacture causal attribution from numerical distance alone.

75. Intervention-Affected Error

Where intervention materially altered the predicted environment, an observed prediction-to-outcome difference may still be measurable.

But it should carry:

Intervention-Affected

qualification.

76. Self-Negating Prediction Error

Example:

Prediction:

90% probability of machine failure

→ preventive maintenance

→ no failure.

A simplistic binary error may record mismatch.

PEMM must preserve intervention context and avoid treating that mismatch as complete performance interpretation.

77. Self-Fulfilling Prediction Error

Where the prediction contributes to creating the predicted outcome, a low observed error may still require downstream qualification.

78. Outcome-Control Context

A predictor that materially controls the outcome may produce extremely low measured error.

The error measurement remains valid.

Its interpretation changes.

Low Error $\neq$ Predictive Skill by Default

79. Predictor–Reality Proximity Context

PEMM carries forward relevant PRP context where material.

Error itself does not reveal whether the predictor was:

public-information; information-advantaged; decision-proximate; outcome-informed; outcome-influencing; outcome-controlling.

80. Revision-Aware Error

Each prediction version may have its own error.

Example:

P1 Error

P2 Error

P3 Error

The final version must not erase earlier version errors.

81. Error Improvement Across Revisions

Error may decrease across successive prediction versions.

This may indicate useful updating.

But interpretation depends upon when and why revisions occurred.

PPCM owns that broader judgment.

82. Last-Minute Error Reduction

A prediction may become highly accurate immediately before outcome resolution.

PEMM records the reduced error.

It does not treat late error reduction as proof of strong early predictive capability.

83. Superseded Prediction Error

A superseded prediction may still have measurable historical error.

Superseded $\neq$ Error-Ineligible

84. Withdrawn Prediction Error

A withdrawn prediction may still have measurable error if a sufficiently governed outcome later becomes available.

Withdrawn $\neq$ Error-Ineligible

85. Decision-Time Error

Where a historical decision relied upon a particular prediction version, PEMM may preserve the error associated with that version.

86. Collective Prediction Error

Collective predictions may have measurable error.

The error record should preserve relevant collective context from CPGS.

87. Prediction Market Error

Market-implied probabilities may support governed event-level probability error measures.

But:

Market Price Error $\neq$ Trading Profit or Loss

88. Cross-Predictor Error Comparison

Errors from different predictors may be compared only where:

Prediction Objects align; Outcome References align; horizons align; measurement methods align; error scales are sufficiently comparable.

89. Cross-Domain Error Boundary

Error values from fundamentally different domains should not be directly compared without legitimate normalization.

5% Error in Domain A $\neq$ 5% Error in Domain B by Default

90. Error History

PEMM preserves historical error states across:

outcome correction; comparison revision; metric revision; prediction-version assessment.

Earlier error records should remain reconstructable.

91. Governance Requirements

A governed PEMM implementation should be able to:

1. consume a Governed Prediction Outcome Comparison;
2. determine Error Eligibility;
3. identify the Error Object;
4. identify all material Error Dimensions;
5. preserve Multi-Dimensional Error;
6. calculate Absolute Error where appropriate;
7. calculate Signed Error where appropriate;
8. distinguish Overprediction from Underprediction;
9. distinguish Absolute from Signed Error;
10. calculate Relative Error where appropriate;
11. calculate Percentage Error where appropriate;
12. identify Zero-Denominator Problems;
13. identify Near-Zero Distortion;
14. preserve Scale Context;
15. support Normalized Error;
16. identify Normalization Basis;
17. preserve Raw and Normalized Error;
18. preserve Numerical Precision;
19. preserve Measurement Precision;
20. identify Rounding Error;
21. preserve Measurement Uncertainty;
22. measure Prediction Range Error;
23. identify Within-Range Outcome;
24. preserve Range Width Context;
25. govern Boundary Error;
26. measure Threshold Error;
27. preserve Threshold Crossing;
28. preserve Threshold Distance;
29. represent Directional Error;
30. measure Magnitude Error;
31. measure Timing Error;
32. distinguish Early and Late Timing Error;
33. identify Horizon Miss;
34. measure Duration Error;
35. represent Sequence Error;

36. represent Categorical Error; 37. prevent arbitrary Category Distance; 38. support Ordinal Error where legitimate; 39. support Binary Error where appropriate; 40. preserve Probability Error Boundary; 41. distinguish Single Probability Error from Calibration; 42. identify Error Measurement Method; 43. govern Method Selection; 44. assess Metric Appropriateness; 45. preserve Metric Consistency; 46. govern Metric Change; 47. prevent Retrospective Metric Change; 48. preserve Error Transformation; 49. identify Error Sign Convention; 50. preserve Error Unit;

51. distinguish Percentage from Percentage-Point Error; 52. preserve Currency Error context; 53. preserve Time-Scale Error; 54. preserve Measurement Transformation Record; 55. support Composite Error; 56. generate an Error Profile; 57. restrict unsupported Aggregate Error; 58. govern Error Weighting; 59. preserve pre-outcome weighting where applicable; 60. prevent Retrospective Weight Manipulation; 61. support Asymmetric Error; 62. distinguish predictive error from operational loss; 63. preserve Tolerance Band; 64. identify Within-Tolerance Error; 65. support Error Materiality classification where governed; 66. preserve Error Materiality Boundary; 67. preserve Error Directionality; 68. identify Systematic Error Candidate; 69. identify Random Error Candidate; 70. preserve Error Cause Boundary; 71. preserve externally governed Error Attribution where applicable; 72. support Intervention-Affected Error; 73. preserve Self-Negating Prediction context; 74. preserve Self-Fulfilling Prediction context; 75. preserve Outcome-Control Context; 76. preserve Predictor-Reality Proximity Context; 77. support Revision-Aware Error; 78. preserve Error Improvement Across Revisions; 79. identify Last-Minute Error Reduction; 80. support Superseded Prediction Error; 81. support Withdrawn Prediction Error; 82. support Decision-Time Error; 83. support Collective Prediction Error; 84. support Prediction Market Error; 85. govern Cross-Predictor Error Comparison; 86. preserve Cross-Domain Error Boundary; 87. preserve Error Qualification; 88. preserve Error State; 89. preserve Error Profile; 90. preserve Error History; 91. preserve Error Measurement Method history; 92. preserve error traceability; 93. produce a Governed Prediction Error Measurement.

92. Governance Logic

PEMM applies the following operational logic:

Has POCM established a Governed Prediction Outcome Comparison?

If no:

→ governed error measurement cannot proceed.

If yes:

→ identify the Error Object.

Is the comparison dimension meaningfully measurable?

If no:

→ classify:

Categorically Representable

Not Meaningfully Measurable

or:

Unable to Determine

If yes:

→ identify the measurement scale.

What type of error applies?

→ numerical;

→ directional;

→ magnitude;

- range;
- threshold;
- timing;
- duration;
- categorical;
- ordinal;
- composite.

What Error Measurement Method is appropriate?

- establish method before calculation.

Are prediction and outcome units aligned?

If no:

- perform governed transformation or restrict measurement.

Does normalization apply?

If yes:

- preserve both Normalization Basis and Raw Error where possible.

Does measurement uncertainty affect the error?

If yes:

- preserve qualification.

Does the prediction contain multiple error dimensions?

If yes:

- create dimension-specific Error Measurements before any aggregation.

Is intervention material?

If yes:

- preserve:

Intervention-Affected

qualification.

Is the prediction revised, superseded, or withdrawn?

If yes:

- preserve version-specific Error Measurement.

Then:

- establish Error State;
- construct Error Profile;
- preserve method, units, qualifications, version, and history;
- pass the governed error record to subsequent POFPS governance.

Canonical transition:

Governed Prediction Outcome Comparison → Dimension-Specific Error Measurement → Governed Prediction Error Measurement

93. Minimum Implementation Framework

A. Prediction Reference

Identifier of the Governed Prediction.

B. Prediction Version

Exact historical prediction version being measured.

C. Comparison Reference

POCM-generated Governed Prediction Outcome Comparison.

D. Error Object

Specific predicted property subject to error measurement.

E. Error Dimension

Value, magnitude, direction, range, threshold, timing, duration, category, sequence, or other governed dimension.

F. Predicted State

Historical predicted value or state.

G. Outcome State

Corresponding sufficiently governed outcome value or state.

H. Error Measurement Method

Method used to quantify or classify the difference.

I. Raw Error

Unnormalized prediction-to-outcome difference where applicable.

J. Normalized Error

Normalized representation where applicable.

K. Error Direction

Overprediction, Underprediction, Early, Late, Directional Match/Mismatch, or other applicable state.

L. Error Magnitude

Measured size or categorical level of the error.

M. Error Qualification

Measurement uncertainty, intervention context, scale limitation, partial comparability, or other qualification.

N. Error State

Measurable, Measurable with Qualification, Partially Measurable, Categorically Representable, Not Meaningfully Measurable, or Unable to Determine.

O. Error History

Versioned historical record of Error Measurements, methods, corrections, and reassessments.

94. Governance Outputs

PEMM may produce:

Governed Prediction Error Measurement Error Eligibility State Error Object Error Dimension Set Absolute Error Signed Error Relative Error Percentage Error Normalized Error Range Error Threshold Error Directional Error Magnitude Error Timing Error Duration Error Sequence Error Categorical Error Ordinal Error Composite Error Profile Error Direction Error Magnitude Error Materiality State Tolerance State Error Measurement Method Error Qualification Intervention-Affected Error State Systematic Error Candidate Error State

Error Profile Error History Error Traceability Record

The primary output is:

Governed Prediction Error Measurement

This becomes governed input for:

PCALM — Prediction Calibration Module

and:

PPCM — Predictive Performance Module

where applicable.

95. Critical Distinctions

Prediction Error $\neq$ Predictive Performance

Prediction Error $\neq$ Methodological Failure

Prediction Error $\neq$ Predictive Capability

Prediction Error $\neq$ Error Cause

No Governed Comparison $\neq$ Governed Error Measurement

One Prediction $\neq$ One Error Value by Default

Absolute Error $\neq$ Signed Error

Percentage Error $\neq$ Universally Valid Error Metric

Same Absolute Error $\neq$ Same Predictive Significance

Normalized Error $\neq$ Raw Error

Observed Value $\neq$ Infinitely Precise Reality

Within Range $\neq$ High Predictive Performance by Default

Horizon Miss $\neq$ Event Non-Occurrence

Different Category $\neq$ Numerical Distance by Default

Single Probability Error $\neq$ Calibration

Available Metric $\neq$ Appropriate Metric

Outcome Knowledge $\neq$ Permission to Select a Better-Looking Error Metric

Error Number Without Unit $\neq$ Complete Error Representation

Multiple Errors $\neq$ Legitimate Single Error Score by Default

Prediction Error $\neq$ Operational Loss

Non-Zero Error $\neq$ Material Error by Default

Measured Error $\neq$ Established Error Cause

Low Error $\neq$ Predictive Skill by Default

Superseded $\neq$ Error-Ineligible

Withdrawn $\neq$ Error-Ineligible

Market Price Error $\neq$ Trading Profit or Loss

5% Error in Domain A $\neq$ 5% Error in Domain B by Default

96. Enterprise Application

PEMM can govern Prediction Error Measurement across:

financial forecasting; economic forecasting; investment analysis; industrial forecasting; predictive maintenance; supply chains; logistics; energy; infrastructure; insurance; project forecasting; technology forecasting; demand forecasting; public policy; environmental forecasting; prediction markets; AI forecasting systems; model ensembles; Human-AI forecasting; autonomous agents; multi-agent prediction systems.

Its function remains constant:

measure how far a prediction was from the governed outcome without pretending that the distance alone explains predictive quality, capability, causation, or value.

97. Use Case 1 — Two Forecasts Miss by the Same Number but Not by the Same Amount

Two forecasting systems produce:

Prediction A: Annual demand = 100 units

Outcome A: Annual demand = 90 units

Absolute Error:

10 units

A second system produces:

Prediction B: Annual demand = 1,000,000 units

Outcome B: Annual demand = 999,990 units

Absolute Error:

10 units

A simplistic error report states:

Both predictions have identical error.

Numerically, their Absolute Error is indeed identical.

But their scale relationship is radically different.

PEMM preserves:

Prediction A

Absolute Error: 10 units Relative Error: approximately 11.1% relative to the observed value, where that denominator is governed as appropriate.

Prediction B

Absolute Error: 10 units Relative Error: approximately 0.001% relative to the observed value.

The critical governance question is:

Does equal numerical distance imply equal predictive error significance?

PEMM answers:

No.

It preserves both:

Raw Error

and:

Normalized / Relative Error

without allowing one representation to silently replace the other.

Same Absolute Error $\neq$ Same Predictive Significance

Normalized Error $\neq$ Raw Error

PPCM may later determine what those errors mean for broader predictive performance.

98. Use Case 2 — Forecast Becomes Accurate Only After Repeated Last-Minute Revisions

A forecasting system predicts completion of a major infrastructure project.

Historical prediction states are:

P1 — Completion: March 2028

Later:

P2 — Completion: June 2028

Later:

P3 — Completion: September 2028

Immediately before completion:

P4 — Completion: 18 September 2028

Actual completion:

20 September 2028

A performance report uses only P4 and states:

Prediction Error: 2 days

PEMM preserves the complete version-specific Error History:

P1 Error: approximately 6 months

P2 Error: approximately 3 months

P3 Error: approximately 20 days, depending on the exact governed date representation

P4 Error: 2 days

The critical governance question is:

Can the final 2-day error legitimately replace the error history of the earlier predictions?

No.

PEMM records each historical prediction state independently.

It does not conclude whether the system performed well or badly.

PPCM later evaluates:

revision trajectory; revision timing; early predictive usefulness; final accuracy; longitudinal performance.

The error history remains:

P1 → Error₁

P2 → Error₂

P3 → Error₃

P4 → Error₄

Final Low Error ≠ Low Error Throughout the Prediction Lifecycle

Revision ≠ Historical Error Replacement

99. Canonical Boundary

PEMM governs the measurement, classification, normalization, qualification, and historical preservation of differences between Governed Predictions and sufficiently governed Outcome References after legitimate Prediction Outcome Comparison. It governs Error Eligibility, Error Objects and Dimensions, Absolute, Signed, Relative, Percentage, Normalized, Range, Threshold, Directional, Magnitude, Timing, Duration, Sequence, Categorical, Ordinal, Composite, asymmetric, weighted, tolerance-aware, intervention-aware, revision-aware,

supersession-aware, and withdrawal-aware error representations; measurement methods, units, transformations, precision, materiality, Error Profiles, qualifications, traceability, and history. It does not establish the Outcome Reference, create Prediction Outcome Association, determine comparison legitimacy, determine probability calibration, assign predictive skill, establish predictive capability, determine error causation, calculate operational value or loss, or determine overall predictive performance.

POAM owns:

Prediction Outcome Association

POCM owns:

Prediction Outcome Comparison

PEMM owns:

Prediction Error Measurement

PCALM subsequently owns:

Prediction Calibration

PPCM subsequently owns:

Predictive Performance

100. Canonical Closing Statement

Every prediction can miss reality by a distance, but that distance has no legitimate meaning until governance establishes what was measured, on what scale, in which direction, under which conditions, and against which historical prediction state.

PEMM makes predictive error measurable without allowing a number to become an explanation, a verdict on methodology, or a substitute for complete performance governance.

Error tells us how far the prediction was from the outcome. It does not, by itself, tell us how good the prediction was.

NON-EXECUTABLE NOTICE

OOF® — OriginOpen® Foundation is a non-executable methodological authority.
This website and all publications are reference-only.

CANONICAL LANGUAGE NOTICE

UCL™ is the canonical language layer of OOF® publications.
English is its carrier language.
In case of ambiguity, the published English HTML text prevails.

RIGHTS & PROTECTION

© OOF® — OriginOpen® Foundation
Protected under MIP® — Methodological Intellectual Property.
All rights reserved.



contact@originopenfoundation.org